```
/*
Name - Ash
Date - 1/11/26
Title - Stop lights with its challenges
 */

const int amarillo = 12;
const int verde = 11;
const int rojo = 13;
const int buzzer = 8;
const int boton = 2;
int Re = 294;
int Do = 262;
int wait = 200;

void setup() {
  pinMode(verde, OUTPUT);
  pinMode(amarillo, OUTPUT);
  pinMode(rojo, OUTPUT);
  pinMode(buzzer, OUTPUT);
  pinMode(boton, INPUT_PULLUP);
}

void loop() {
  digitalWrite(verde, HIGH);

bool botonPresionado = false;

  for (int i = 0; i < 40; i++) {
    if (digitalRead(boton) == LOW) {
      botonPresionado = true; // Guardamos que se presionó el botón
```

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    break; // Rompemos el ciclo al instante
  }
  delay(100);
}

digitalWrite(verde, LOW);

if (botonPresionado) {
  digitalWrite(rojo, HIGH);
  tone(buzzer, Re);
  delay(3000);
  noTone(buzzer);
  digitalWrite(rojo, LOW);
  delay(2000);
}

else {

digitalWrite(amarillo, HIGH);
tone(buzzer, Do, wait);
delay(200);
digitalWrite(amarillo, LOW);
delay(1000);



digitalWrite(amarillo, HIGH);
tone(buzzer, Do, wait);
delay(200);
digitalWrite(amarillo, LOW);
delay(1000);
```

```
  digitalWrite(amarillo, HIGH);
  tone(buzzer, Do, wait);
  delay(200);
  noTone(buzzer);
  delay(200);
  digitalWrite(amarillo, LOW);


  digitalWrite(rojo, HIGH); // (5) Variable del pin rojo
  tone(buzzer, Re);
  delay(3000);
  noTone(buzzer);
  digitalWrite(rojo, LOW);
  delay(4000);
  }
}
```